

Assessing DHEA and Cortisol Diurnal Patterns

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Introduction

Research suggests that there is a relationship between hormonal patterns across the span of a day and conditions including sleep disorders and depression. In this study, the focus is on the hormones DHEA and cortisol, which play an important role in regulating the body's stress response. Cortisol is an important steroid hormone that significantly affects metabolism, which in turn influences the central nervous system. Cortisol typically follows a diurnal pattern, with levels peaking shortly after waking and then gradually declining throughout the day. DHEA is another steroid hormone involved in the stress response and is often considered to have effects that counterbalance cortisol. DHEA has been associated with immune function, mood regulation, and overall physiological resilience, and its daily pattern is less well characterized compared to cortisol. Because both hormones are involved in stress regulation, understanding their patterns across the day may provide insight into physiological processes related to health and disease.

In this study, researchers examined cortisol and DHEA patterns over the course of a day across a total study period of three days. Saliva samples were collected using two methods: a Saliva Procurement and Integrated Testing (SPIT) booklet and an automatic electronic monitoring cap. Samples were collected at waking, 30 minutes after waking, before lunch, and 600 minutes after waking. Subjects were allowed to follow their normal daily schedules and were not required to adhere to a specific wake-up time. A total of 31 healthy control subjects participated in the study.

This report addresses the following hypotheses of interest:

Hypothesis 1: There was satisfactory agreement between the subject's recording of sampling times compared to the recording times by the electronic monitoring cap. More

specifically, there is a one-to-one relationship between the subject's recorded times and the booklet times.

Hypothesis 2: Subjects adhered appropriately to the study protocol of the +30 minute and +600 minute sampling times.

Hypothesis 3: DHEA and Cortisol levels show an initial increase from waking to 30 minutes followed by a gradual decline from 30 minutes and on.

Note: We will further refer to the electronic monitoring cap as MEM time and the the SPIT Booklet as Booklet time for the remainder of the report.

Methods

Preliminary Data Analysis

Below is a Table1 of the primary variables of interest for the analysis. The minutes are all minutes relative to midnight.

Table 1: Summary Per Subject Within Collection Sample

	Wake Time	+30 Min	Pre Lunch	+600 Min
n	31	31	31	31
Booklet Minutes (mean (SD))	414.00 (65.38)	445.91 (65.84)	735.34 (51.23)	1056.11 (107.27)
MEM Minutes (mean (SD))	415.61 (67.62)	482.20 (86.13)	734.24 (40.46)	1065.95 (110.96)
Minutes Since Wake (mean (SD))	0.00 (0.00)	33.38 (4.89)	324.04 (69.82)	645.48 (79.41)
DHEA (nmol/L) (mean (SD))	1.78 (0.94)	1.11 (0.73)	0.53 (0.46)	0.50 (0.60)
Cortisol (nmol/L) (mean (SD))	7.38 (4.47)	9.01 (3.56)	3.31 (1.91)	4.08 (7.61)

For analysis, we exclusively worked with the time variables in minutes. Looking at Minutes Since Waking, we can see that the mean minutes closely follow the study protocol times. The average DHEA and Cortisol levels appear to be similar across the 4 time points throughout the day.

As missingness can present serious issues for the data analysis if not addressed, we investigated missingness through calculating the number and percentage of missing values

for each variable in the dataset. We further visualized missingness for each variable.

For the variables we employed in our analyses, we assessed and visualized distributions to determine if there were any outliers or considerations that needed to be taken for data analysis.

Data Cleaning

For DHEA, we removed observations that were ≥ 5.205 (nmol/L) as these are considered extremely high DHEA levels that could interfere with analysis. There were 6 observations (4 of which corresponded to Subject 3037) that had DHEA levels of 5.205 (nmol/L). We removed all observations for Subject 3037 due to multiple high DHEA values that suggest underlying health problems as well as the 2 other observations with DHEA of 5.205 (nmol/L). For cortisol, we observed 5 observations from 4 patients that had Cortisol levels ≥ 26 . Although these values are considered high, they are biologically plausible and they were left in the analysis. One observation had a cortisol level of ~89 which was removed for the analysis.

Data Analysis

To address the first hypothesis, we employed a longitudinal linear mixed effects model with Booklet Time in Minutes as the outcome with covariates - MEM in Minutes and the Day Number (1-3). Adding Day as a covariate noticeably improved diagnostic plots for the assumptions of heteroscedasticity and normality. We included a random intercept for subject that allowed us to take variability between subjects into account in our model. To assess whether booklet times tended to be recorded earlier or later than MEM time, we used the difference between MEM and Booklet times as the outcome in a longitudinal linear mixed effects model. This model had Day Number as a covariate and a random intercept for subject.

To determine whether subjects were adhering accurately to the +30 minute and +600 minute sampling times required by study protocol, intervals of time were defined to cal-

culate percentages for which there was good or adequate adherence. Good adherence was defined by a collection of the saliva sample within a 7.5 minute window interval. Patients had adequate adherence if samples were within a 15 minute window. To calculate the percentage of individuals that had good adherence and adequate adherence, the difference between the “correct” sampling time and the observed time was calculated and this difference was used to assess whether observed times were within the good or adequate adherence windows. We visualized and summarized the distribution of the MEM and Booklet times to guide us in our decision to proceed with either MEM or Booklet times for our calculations. We observed that MEM times had a considerable amount of missingness and determined that reporting both would give a better overall sense of adherence. We summarized adherence for the study population with tables.

For the third hypothesis, we used a piecewise longitudinal linear mixed effects model to assess whether DHEA and Cortisol levels followed the expected diurnal pattern. Due to a right-skew in the distribution of DHEA, a log transformed DHEA variable was used as the outcome to assess DHEA changes over the day. No transformations was made to Cortisol. Covariates for both models included manually calculated pre and post 30 minute variables (variables for piecewise slopes) which would give us the slopes for the two time intervals of interest.

All analyses were conducted in R version 4.4.1 (2024-06-14), using the nlme package for modeling.

Results

From the missingness investigation, we determined the MEMs had the highest amount of missing: 177 missing values which is equivalent to 47.6%. Booklet had the second highest amount of missingness with 55 missing values or 14.8%. Cortisol and DHEA measurements had 5 missing values each or 1.34% missingness.

Hypothesis 1

Below is a table of the estimates, 95% Confidence Intervals and P-values:

Table 2: Model Summary

	Value	Std.Error	DF	t-value	p-value
(Intercept)	3.436	4.355	41	0.789	0.435
MEM_C	0.986	0.010	41	102.912	0.000
factor(DAYNUMB)2	-3.148	1.550	41	-2.030	0.049
factor(DAYNUMB)3	-1.642	1.558	41	-1.053	0.298

Holding day constant, a one-minute increase in MEM-recorded wake time is associated with an approximately one-minute increase in Booklet-recorded wake time ($\beta \approx 0.99$; 95% CI: (0.97–1.01); $p < 0.0001$). This near one-to-one association indicates strong concordance between the two wake-time measures within a given day. In other words, the subject's recorded wake up time and the MEM's recorded wake up time track each other quite closely. This is visualized in Figure 1 below. Changes in MEM time are mirrored almost exactly by changes in Booklet time and our initial hypothesis is correct. In regard to bias, booklet times tended to be recorded approximately 7-8 minutes earlier (95% CI: (-14.28, -0.62)) than MEM times on average. This difference is statistically significant with $p\text{-val} \approx 3$.

Hypothesis 2

Below we summarize the percentage of subjects that adhered to the study sampling times, for both MEM and Booklet times.

Table 3: Percentages for Good and Adequate Adherence by Collection Window

Window	Method	Good	Adequate
+30 minutes	Booklet	78.16	89.66
+30 minutes	MEM	52.86	71.43
+60 minutes	Booklet	46.25	56.25
+60 minutes	MEM	32.53	39.76

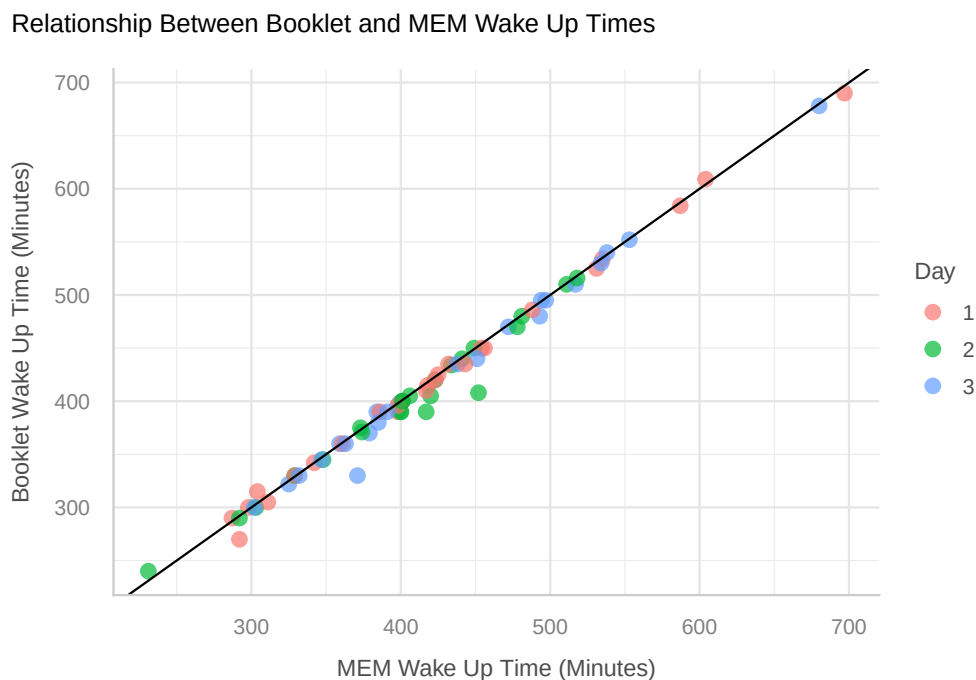


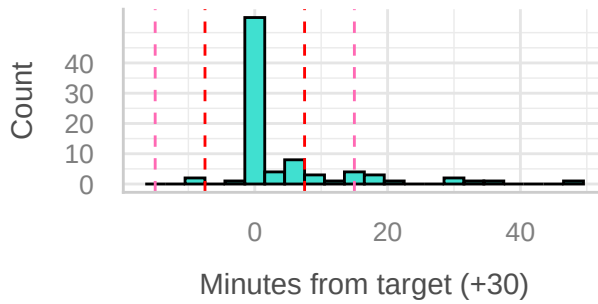
Figure 1: Booklet Vs. MEM Wake Up Times

Overall, more subjects are meeting the adequate adherence thresholds compared to the good adherence thresholds for both the +30 minute and +600 minute sampling times. In addition, the MEM percentages are lower for both sampling times and for both good and adequate thresholds.

In Figure 2 we visualize the minutes from the +30 minute sampling time (left column) and the +600 minute sampling time (right column). Times for Booklet are in the top row and Times for MEM are in the bottom row. The red dashed lines indicate the ± 7.5 window for good adherence while the pink dashed lines are the ± 15 minute window for adequate adherence.

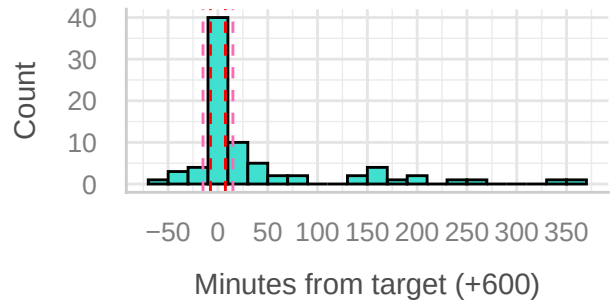
Minutes from +30 Minute Sampling Time

Booklet



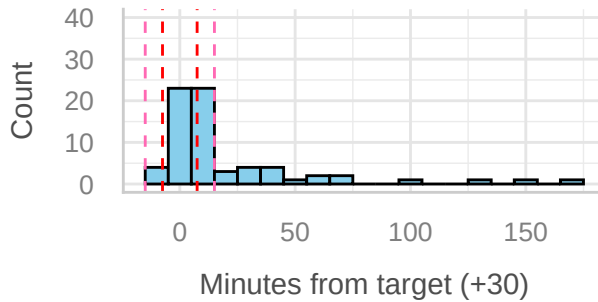
Minutes from +600 Sampling Time

Booklet



Minutes from +30 Minute Sampling Time

MEM



Minutes from +600 Minute Sampling Time

MEM

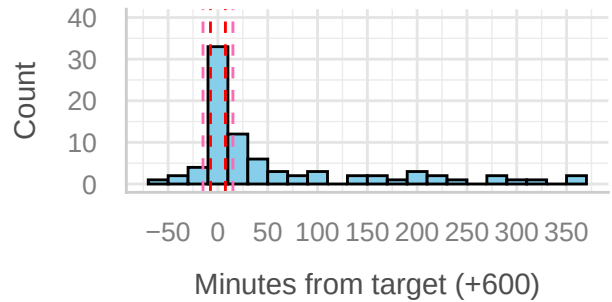


Figure 2: Adherence for +30 and +600 Sampling Times

Hypothesis 3

DHEA

In Table 3 we display the estimates, 95% Confidence Intervals and P-values for the DHEA model:

Table 4: Model Summary

	Value	Std.Error	DF	t-value	p-value
(Intercept)	0.21392	0.11741	288	1.82195	0.0695
time_pre30	-0.01724	0.00301	288	-5.73083	0.0000
time_post30	-0.00153	0.00016	288	-9.87455	0.0000

For the time period of wake time to 30 minutes, there is approximately a 40% decrease in

DHEA (95% CI: (-50.04%, -28.75%)). This is statistically significant with $p < 0.001$. DHEA shows a statistically significant decrease, not the expected increase, from waking to 30 minutes after waking.

As hypothesized, from 30 minutes to 600 minutes, there is approximately a 60% decrease in DHEA levels (95% CI: (-66.65%, -51.91%)). This is statistically significant with $p\text{-val} < 0.001$. We visualize these slopes in Figure 3 below.

DHEA Pattern Over Time

Knot at 30 minutes

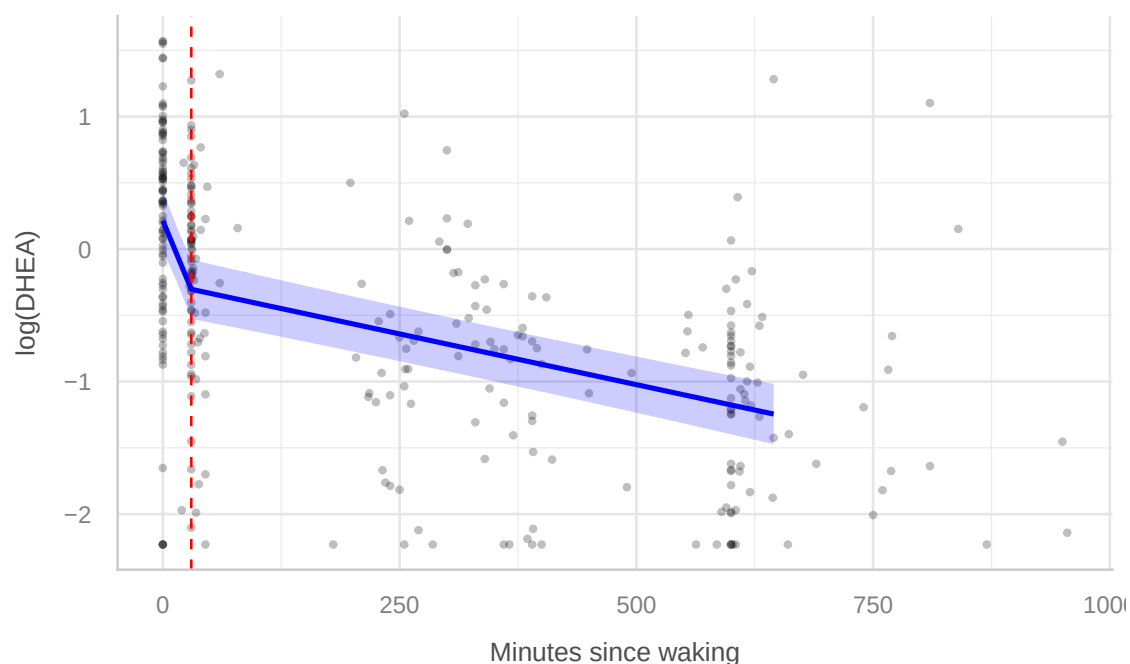


Figure 3: Diurnal DHEA Pattern

Cortisol

For the time period of wake time to 30 minutes, there is approximately a 54% increase in Cortisol (95% CI: (-60.19%, 503.15%)). This is not statistically significant with $p = 0.53$. Although there is an increase from waking to 30 minutes after waking, it is not a statistically significant increase.

Table 5: Model Summary

	Value	Std.Error	DF	t-value	p-value
(Intercept)	7.22161	0.56345	287	12.81668	0.00000
time_pre30	0.02522	0.02295	287	1.09886	0.27275
time_post30	-0.00938	0.00119	287	-7.91625	0.00000

From 30 minutes to 600 minutes, there is approximately a 99% decrease in DHEA levels (95% CI: (-99.86%, -97.27%)). This is statistically significant with $p\text{-val} < 0.001$. As expected, Cortisol showed a statistically significant decline over 10 hours. Cortisol follows the expected pattern of an increase (although not significant) from waking to 30 minutes followed by a decline from 30 minutes and on.

Cortisol Pattern Over Time

Knot at 30 minutes

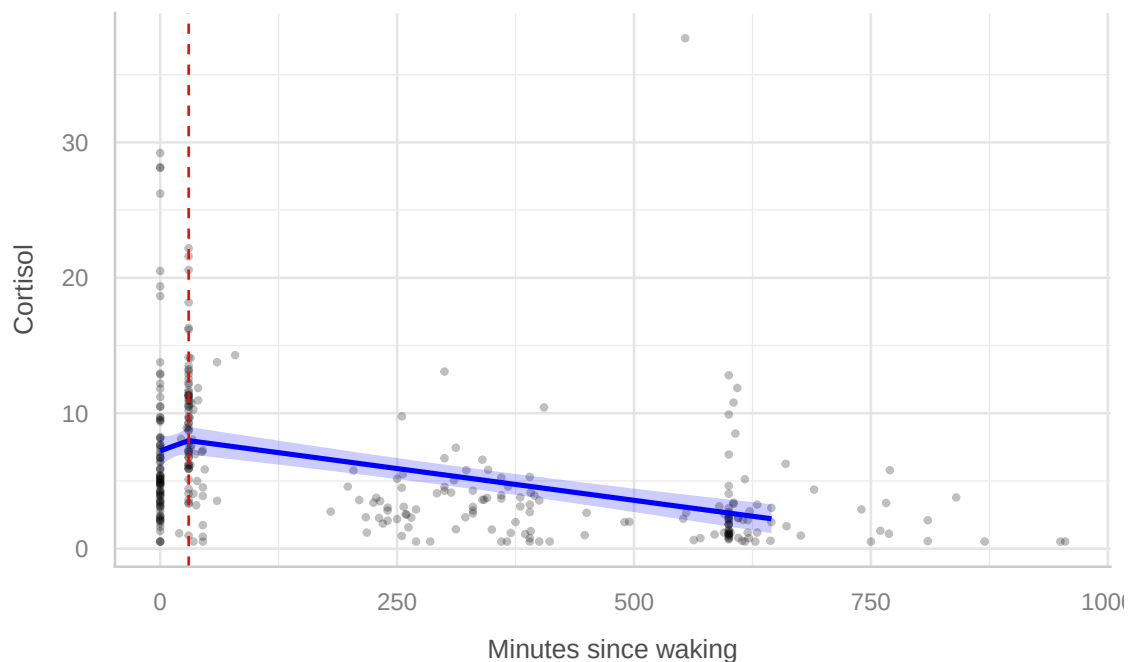


Figure 4: Diurnal Cortisol Pattern

Discussion and Summary

In regard to the comparison of MEM and booklet times, we found a statistically significant and strong linear relationship between the two measures, indicating that booklet-recorded times tracked closely with MEM times. However, we also observed that booklet times tended to be recorded slightly earlier than MEM times, on average. This suggests the presence of a small systematic bias. Possible explanations include delayed electronic cap recordings due to technical issues or perhaps inaccuracies in self-reported booklet times. This highlights a known limitation of studies relying on self-reported sampling times.

For the +30 minute sampling window, the percentages for both good and adequate adherence were reasonably high, at approximately 80%. On the other hand for the +600 minute sampling window, we saw a notable decrease in adherence. Both good and adequate adherence percentages for booklet and MEM times fell below 50%. This decrease is expected, since longer intervals from waking are more susceptible to schedule variability which undoubtedly can affect accurate timing. Across both sampling windows and adherence thresholds, the impact of missing MEM time was apparent, with lower adherence percentages compared to booklet time.

In the assessment of DHEA and cortisol patterns over time, we found that cortisol followed the expected diurnal pattern, with higher levels earlier in the day and a decline over time. In contrast, DHEA did not exhibit a clear or consistent pattern over the sampling period. These differences may be due to several factors, including biological differences in hormone regulation, variability within this particular cohort, or limitations in sampling frequency and timing.

In summary, our findings suggest that booklet-based saliva collection using the SPIT device demonstrates strong agreement with electronic monitoring, though a small systematic bias toward earlier recorded booklet times was observed. Adherence to short-term sampling requirements was generally high, while adherence to longer-term sampling windows

was more variable. Cortisol exhibited the expected diurnal decline, whereas DHEA did not show a clear temporal pattern in this sample. These results highlight the feasibility and limitations of at-home saliva collection protocols and emphasize the importance of accurate timing and careful handling of missing data in studies of diurnal hormone patterns.

Data and Code Availability

The data and statistical code used to generate the results and figures in this report are available in the following GitHub repository:

- GitHub repository: <https://github.com/DianaF65/BIOS6624>

Directory structure:

- Background/ - A description of the background of the project.
- Code/ - A R Markdown file consisting of code for exploratory and an initial report draft as well as as a .R file of code for the complete analysis.
- RawData/ – Raw data files.
- Reports/ R Markdown file used to generate this report

All analyses and figures were generated using R, and the repository contains the R scripts and R Markdown files used to produce this report